

API Reference

JSPL EAF Industrial AI Decision Support — Release 1.0
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JSPL EAF TTT REST API — Release 1.0

Base URL: ``http://localhost:8001``

Interactive docs: ``/docs`` | OpenAPI: ``/openapi.json``

Version: 1.0.0 (Release 1.0 — Industrial Demonstration Ready)

Endpoints

	Method	Path	Description
----- ----- -----			
	GET	<code>`/health`</code>	Service health, model/optimizer load status, version registry
	GET	<code>`/version`</code>	Frontend/backend/model/optimizer/research version registry
	GET	<code>`/model-info`</code>	Phase 19 model metadata and features
	POST	<code>`/predict`</code>	Predict TTT with CI, warnings, and explainability
	POST	<code>`/optimize`</code>	Phase 20.2 physics-guided recipe optimization
	POST	<code>`/optimize/v2`</code>	Phase 31 Optimizer V2 (research – never optimizes POWER)
	POST	<code>`/hybrid/evaluate`</code>	Phase 32 hybrid trust framework
	POST	<code>`/whatif`</code>	Tornado sensitivity analysis
	GET	<code>`/historical`</code>	Historical comparison (default recipe)
	POST	<code>`/historical`</code>	Historical comparison + distributions
	GET	<code>`/historical/statistics`</code>	Full min/P5/median/P95/max/mean/std
	POST	<code>`/process-health`</code>	Process health gauges
	GET	<code>`/report?format=json\ csv\ pdf`</code>	Report for default recipe
	POST	<code>`/report`</code>	Report for custom recipe
	GET	<code>`/validation`</code>	Plant validation history + MAE/RMSE
	POST	<code>`/validation`</code>	Record plant validation result
	GET	<code>`/feedback`</code>	Operator feedback history
	POST	<code>`/feedback`</code>	Submit operator feedback
	GET	<code>`/feedback/summary`</code>	Acceptance rate summary
	GET	<code>`/reliability/summary`</code>	Reliability dashboard aggregates
	GET	<code>`/deployment/readiness`</code>	Traffic-light readiness assessment

Default recipe

```
```json
{
 "HM": 56.8, "DRI": 63.2, "HBI": 0, "Bucket": 0,
 "LIME": 9.9, "DOLO": 2.5, "CPC": 576,
 "POWER": 29985, "OXY": 3911,
 "Shift": "B", "Power_Restriction": 0
}
```
```

Expected prediction: **~39.9 min** (120 t total charge).

Industrial validation

- Total charge is never a hard rejection criterion.
- Out-of-band inputs return ``validation_warnings`` and ``confidence`` instead of HTTP 422.

- Request field `POWER` = Electrical Energy (kWh) in responses.

Explainability (Phase 29)

Optional `explainability` on `/predict` and `/optimize` — no additional ML inference.

`POST /predict` — explainability fields

`similar\_heats`, `contributor\_interpretations`, `prediction\_quality`, `industrial\_observations`,
`digital\_twin\_readiness`, `historical\_similarity\_pct`, `industrial\_risk`

`POST /optimize` — explainability fields

`validated\_recommendations`, `recommendation\_confidence`, `recommendation\_stability`,
`top5\_alternatives`, `recommendation\_narrative`, `penalty\_breakdown`, `similar\_heats`,
`digital\_twin\_readiness`

Phase 31 V2 (`POST /optimize/v2`)

- `power\_optimized`: always `false`
- `recommendations[]`: top 5 with `physics\_score`, `industrial\_score`, `explanation`
- Does not replace `/optimize` (Phase 20.2)

Phase 32 Hybrid (`POST /hybrid/evaluate`)

- `reliability\_index`, `ai\_confidence`, `physics\_confidence`, `industrial\_confidence`
- `historical\_similarity\_pct`, `recommendation\_stability`, `consensus`, `decision\_tree`

Phase 33 validation

- `POST /validation`: record heat_number, predicted_ttt, actual_ttt, optimizer_used
- `GET /validation`: returns entries + metrics (MAE, RMSE, bias, MAPE)

Version registry (`GET /version`)

```
```json
{
 "frontend_version": "1.0.0",
 "backend_version": "1.0.0",
 "research_phase": "Phase 33 (frozen)",
 "model_phase": "Phase 19",
 "optimizer_phase": "Phase 20.2"
}
```
```

PDF export

See `release/API\_REFERENCE.pdf` for printable reference.